

Serway and Jewett - Physics for Scientists and Engineers

Chapter 4: Motion in Two Dimensions

4.1 The position, velocity, and acceleration vectors of a particle moving in a plane.

### 4.3 Projectile Motion

A projectile moves with constant horizontal velocity and constant downward acceleration  $g$ . Its path is a parabola. The horizontal range of a projectile launched with speed  $v_i$  at angle  $\theta$  is  $R = v_i^2 \sin(2\theta) / g$ , so the maximum range is reached when  $\theta$  is 45 degrees. The maximum height is  $h = v_i^2 \sin^2(\theta) / (2g)$ .